

Test Yourself

Microeconomics

Price Elasticity of

Supply

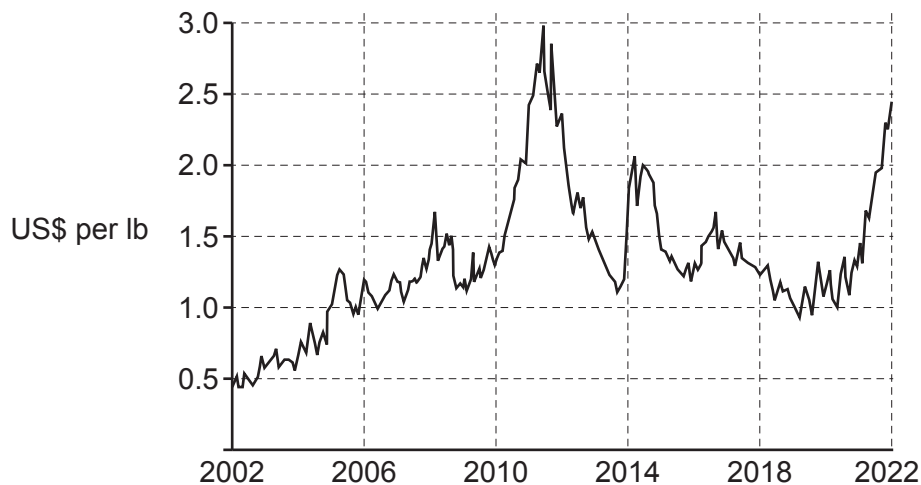
Section A

Case Study 1dy 1

23/O/N/244

Global coffee bean prices reach a new high

Coffee is big business, especially in the United States (US). The global market was worth nearly US\$110bn in 2020, with production around 10m tonnes of coffee beans. 95% of this production came from thousands of small-scale producers in South America, Central America, Asia and Africa. If present trends continue, production is forecast to triple by 2050. The Arabica coffee bean price was at a 10-year high in January 2022, having more than doubled in 2021. Fig. 1.1 shows the world price of coffee in United States dollars (US\$) per pound (lb) weight. A pound is 454 grammes.



Source: *tradingeconomics.com*, 18 February 2022

Fig. 1.1: Arabica coffee bean price, 2002 to 2022

So how can this huge price increase be explained? The main cause was a series of weather events affecting Brazil, the world's largest producer of high quality Arabica coffee beans. Its market share is 35% of total global supply. There was severe drought in early 2021 that reduced the number of 'cherries', which contain the beans on coffee bushes. The crop yield was further damaged by frosts that followed the drought. As a result of these weather events, Brazil produced its smallest volume of quality coffee beans for ten years.

In addition, the production of cheaper low-quality Robusta coffee beans in Vietnam was badly affected by storms which stripped the bushes of their 'cherries'. Overall, there was a very large decrease in the global supply of all coffee beans.

Supply issues affecting major producing countries like Brazil and Vietnam mean that the volume of coffee beans produced regularly fluctuates between 'high' years and 'low' years. Producers try to reduce their risks through buffer stock schemes in order to maintain a regular income stream. This type of scheme is particularly important for the many small-scale subsistence producers who have no other source of income.

The dramatic weather events of 2021 have been exceptional, although a few large-scale producers in Brazil who survived the immediate impact of drought and frost should gain from the huge rise in the price of coffee beans. But what about the thousands of other producers elsewhere who lack power in the market? The most likely outcome is that they will once again become victims of the unpredictable global market for coffee beans.

A global olive oil shortage

Almost every kitchen in the United States (US) will pay the price for the intense heat and drought in Europe in the summer of 2023. Olive harvests in key countries were so poor that some analysts were concerned that there would be a shortage of olives to produce enough olive oil to meet the demand around the world. The dramatic reduction in output of this typical household cooking oil has resulted in the consumer price of olive oil rising between 30% and 50% across the US. At the same time, the US dollar has fallen in value against Europe's main currency (the euro), further adding to the price increase of olive oil.

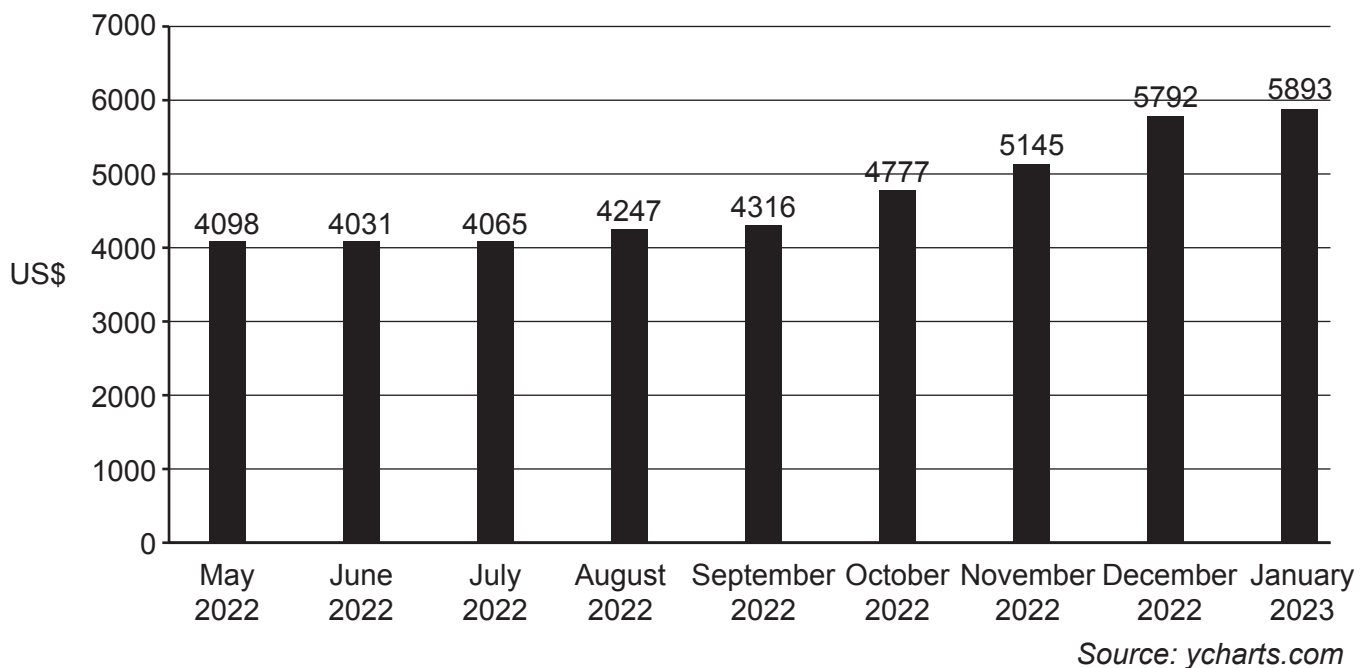
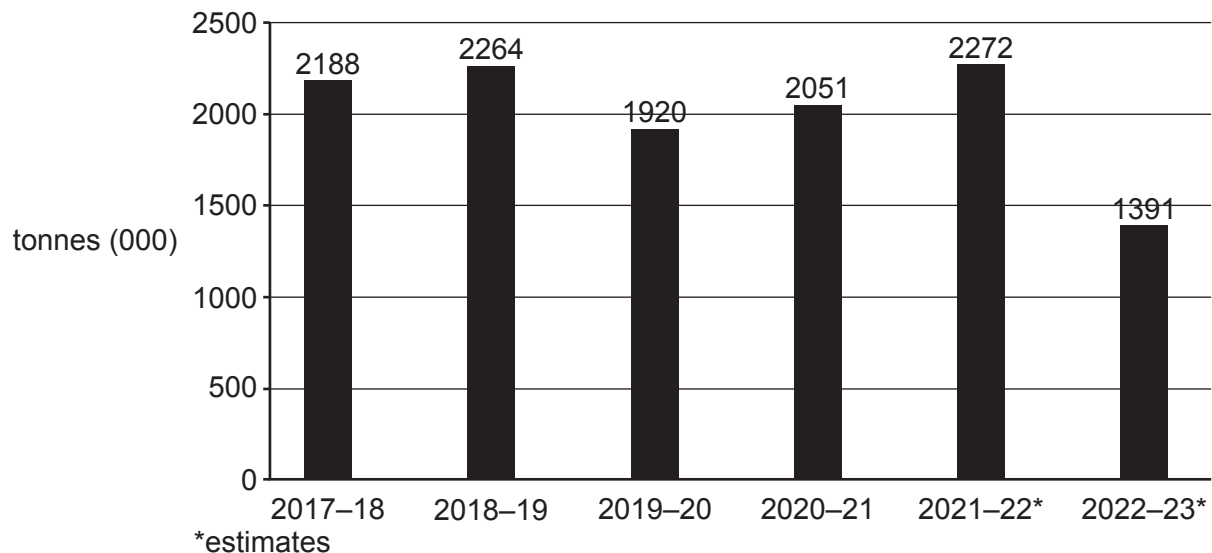


Fig. 1.1 World price of olive oil, May 2022 to January 2023 (US\$/tonne)

In Spain, olive production was forecast to be down by about 50% in 2023, which would make it the lowest since 2008. The wide-scale drought conditions across Southern European and other Mediterranean countries significantly impacted on the global supply of olives, since about 80% of global output comes from these regions.



Source: DG Agri olive oil dashboard

Fig. 1.2 European Union (EU) production of olive oil, 2017 to 2023

Restaurants and commercial kitchens are desperately trying to find a solution to the olive oil shortage, but alternatives are scarce too. Sunflower oil is a top choice for an olive oil replacement, but the conflict in Ukraine, which is the biggest producer of sunflower oil, has made this substitute increasingly difficult to access.

Previously, the US would look to other countries for olive oil, but supply chains have been severely disrupted. Olive oil does not store very well as its quality deteriorates over time, so increasing stocks does not solve the problem. It is perhaps unsurprising that analysts are recommending investment into new methods of storage that will preserve the quality of the olive oil. This would allow stocks to increase after good harvests to help stabilise prices during future periods of poor harvests.

Source: Adapted from: *Olive oil prices climbing after heat, drought in Europe leads to poor harvest*, Foxweather, 21 February 2023

Case Study 1

- (a) Use a demand and supply diagram to show why there was a ‘huge’ increase in the price of Arabica coffee beans in 2021. [2]
- (b) Is the price elasticity of supply of coffee beans likely to be elastic or inelastic in any one year? Justify your answer from the information provided. [2]
- (c) Assess the extent to which coffee bean producers will gain from the huge increase in coffee bean prices in 2021. [6]

Case Study 2

- (a) Using Fig. 1.1, calculate the percentage change in the world price of olive oil between May 2022 and January 2023. [2]
- (b) With reference to Fig. 1.2 and the help of a demand and supply diagram, consider the possible impact of increased olive oil prices in 2023 on consumer surplus. [4]
- (c) Assess the likely impact of increasing stocks of olive oil on the price and quantity of olive oil traded in the future. [6]

Section B

Duration of each question is **40 mins**

1. With the help of a diagram, explain what could cause an increase in demand for a product and consider whether the impact of an increase in demand on the price of the product will be the same in the short run and the long run. (8) **q2a/21/O/N/23**
2. Assess whether the concept of price elasticity of supply or cross elasticity of demand will be more useful to a business wanting to increase its total sales in a growing economy. (12) **q2b/22/O/N/23**
3. The price elasticity of supply (PES) for a new smartphone is estimated to be 0.8 in the short run and 1.8 in the long run.
Explain what these estimates mean for producers and consumers of smartphones and consider why the estimates differ. (8) **q2a/23/O/N/23**
4. Assess whether the supply of agricultural products is likely to be more price elastic or less price elastic than the supply of manufactured products. (12) **q2b/21/O/N/24**
5. Assess whether an estimate of the price elasticity of demand for a product is likely to be more useful to a firm than an estimate of its price elasticity of supply. (12)
q3b/22/F/M/24
6. Assess the extent to which price elasticity of supply or cross elasticity of demand is most useful to businesses. (12) **q3b/21/M/J/24**

7. With the help of a formula, explain the meaning of price elasticity of supply **and** consider which factors are most important in determining whether the price elasticity of supply for a good is likely to be relatively elastic or relatively inelastic. (8) **q3a/specimen23**

8. Assess the likely success of any **two** government policies to make the price elasticity of supply of all agricultural products more elastic. (12) **q3b/specimen23**

Section C

Duration 1 hour

1. When the price of good X is \$2.00, the quantity supplied is 100 000. The price elasticity of supply of good X is 0.8 in the short run and 1.4 in the long run. The price of good X increases to \$2.20.

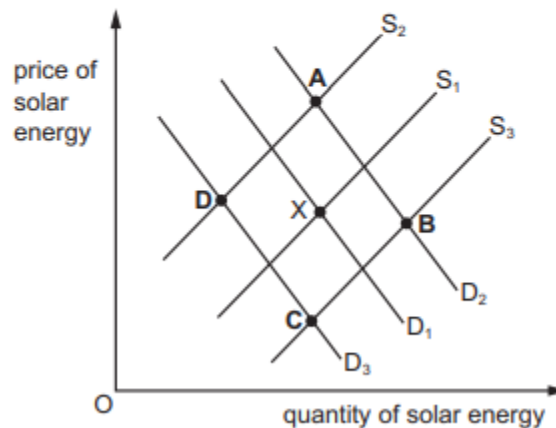
What is the increase in the quantity supplied of good X between the short run and the long run?

- A 6000 B 60 000 C 114 000 D 140 000

2. The diagram shows the demand and supply curves for solar energy in India. The initial equilibrium point is at X.

The Indian government increases the subsidy to solar energy producers. At the same time, there is a rise in the price of alternative sources of energy.

What is the new equilibrium position?

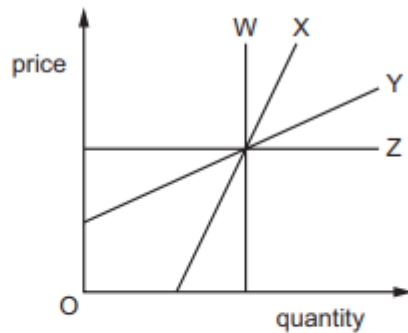


3. What is most likely to cause a shift in the supply curve for rice?

- A a change in the price of the rice
B a change in the price of meat
C a change in the size of the rice industry
D a change in tastes and preferences for rice

4. What does joint supply mean?
- A A firm can choose between producing a range of different goods.
 - B A good has multiple uses.
 - C A good is a by-product of the production process of another good.
 - D Consumers consider two goods as complements.

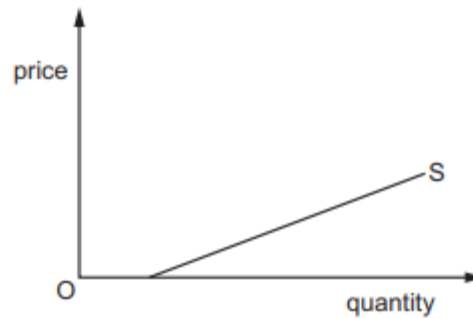
5. The diagram shows four supply curves.



Which statement about the price elasticity of these supply curves is correct?

- A W has elasticity of 0 that will rise as price rises.
 - B X has elasticity greater than 1 that will be constant as price rises.
 - C Y has elasticity greater than 1 that will fall as price rises.
 - D Z has elasticity of 0 that is constant as quantity rises.
6. Product X has a price elasticity of supply (PES) of +2, whilst product Y has a PES of +0.2.
- Which statement about products X and Y is correct?
- A X has more substitutes than Y.
 - B A 20% price fall would lead to a greater decrease in production of Y than of X.
 - C After a price rise, it is more difficult to increase Y's output than X's.
 - D After a price fall of 10%, more people would buy X than would buy Y.

7. The diagram shows the supply curve of a firm.



Which statement about the price elasticity of supply (PES) along this curve is correct?

- A PES is greater than unity at all points along the supply curve.
 - B PES is the same at all points along the supply curve.
 - C PES is elastic when the price is low and becomes more inelastic when the price rises.
 - D PES rises when the supply rises.
8. The supply, S , of a product is determined by the equation

$$S = 10 + 10P$$

where P is the price of the product in \$.

What is the product's price elasticity of supply when its price rises from \$1 to \$2?

- A 0.5
 - B 1.0
 - C 2.0
 - D 5.0
9. The supply curve for a firm is a horizontal straight line.

What can be concluded from this statement?

- A Quantity supplied is infinite at the given price.
- B Quantity supplied is infinite below the given price.
- C Quantity supplied remains constant at all prices.
- D Quantity supplied is perfectly inelastic.

10. Butter and buttermilk are goods in joint supply. Supply curves in both markets slope upwards to the right. Health concerns cause people to reduce their consumption of butter.

What are the initial consequences of this for supply in the markets for butter and buttermilk?

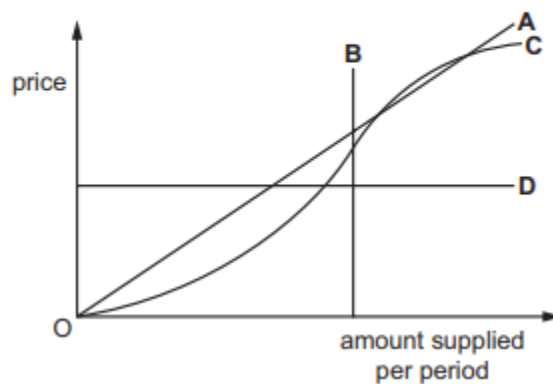
	supply of butter	supply of buttermilk
A	move left along supply curve	move left along supply curve
B	move left along supply curve	shift left of the supply curve
C	shift left of the supply curve	move left along supply curve
D	shift left of the supply curve	shift left of the supply curve

11. What most accurately describes a market supply curve?

- A** supply at different income levels, assuming product prices remain unchanged
- B** supply at different levels of factor prices, assuming product prices remain unchanged
- C** supply at different prices, assuming no changes in technology
- D** supply at different time periods, assuming no changes in technology

12. The diagram shows four supply curves.

Which curve has a price elasticity of supply of 1 for all levels of quantity supplied?



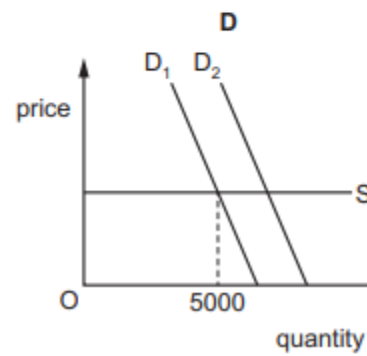
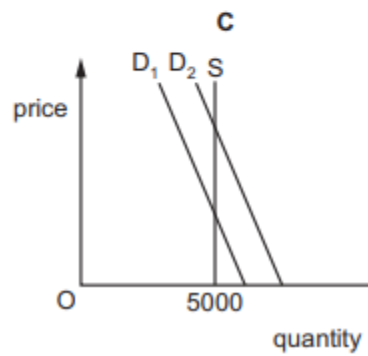
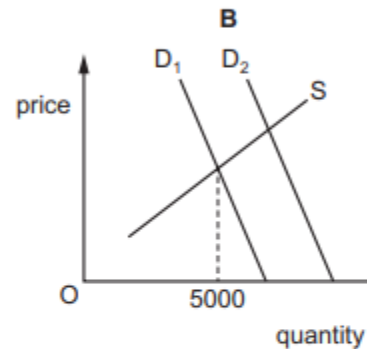
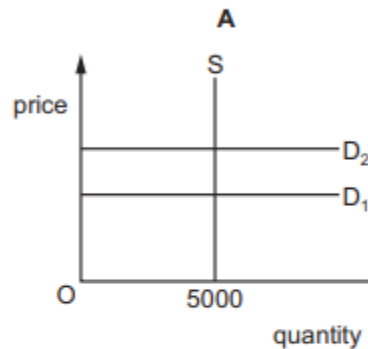
13. The supply, S , of a product is determined by the equation $S = 10 + 10P$, when P is the price of the product.

What is the price elasticity of supply when the price changes from \$1 to \$2?

- A** 0
- B** 0.5
- C** 1.0
- D** 2.0

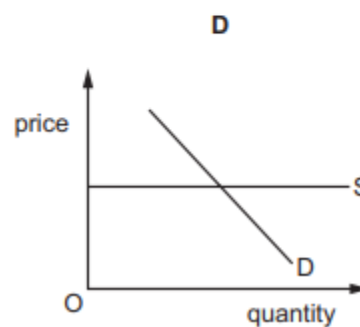
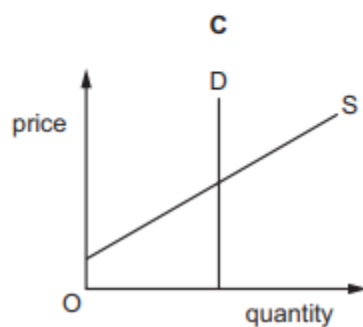
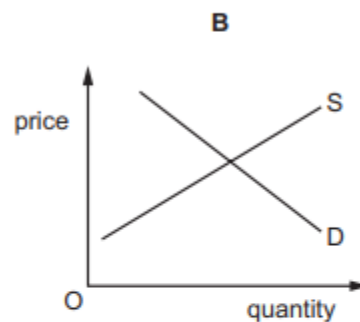
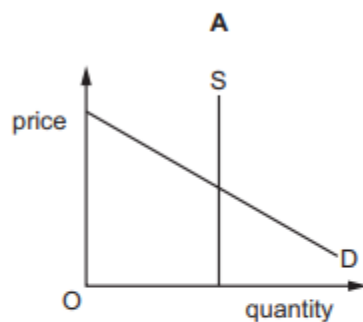
14. A popular band is due to perform at a music concert in a venue that has a 5000-seat capacity. Recent appearances and the release of a new album have made the band even more popular. D_1 and S represent the original demand and supply curves for concert tickets and D_2 the new demand curve.

Which diagram best represents the likely outcome on the market for concert tickets?



15. The diagrams relate to the markets for four different products.

In which market would a 10% decrease in demand and simultaneously a 10% increase in supply definitely result in both a fall in equilibrium price and a rise in equilibrium quantity?



16. The supply of a new drug to relieve muscle pain is very inelastic over a short period of time. The supply of printed newspapers is much more elastic over the same time period.

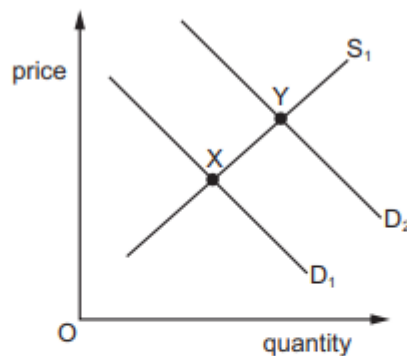
What could explain the difference in elasticity of supply of the drug and printed newspapers in this time period?

	new drug	printed newspapers
A	cost per unit is low	cost per unit is high
B	discovery is by chance	takes longer to produce
C	research and development essential	excess capacity in printing
D	similar products available	similar products unavailable

17. In which situation is the price elasticity of supply for a product most likely to be relatively low?

- A Demand for the product comes from a wide range of customers.
- B Producers in closely related industries can easily switch to making the product.
- C Manufacture of the product requires highly skilled labour.
- D The main raw material used in the production of the product is in abundant supply.

18. The diagram shows the market for electric steel-plated switches used in new houses.



What could **not** have caused the market equilibrium position to have changed from X to Y?

- A increasing the availability of loans for buying houses
- B granting a subsidy for producers of bricks
- C removing sales tax on electric switches
- D removing tariffs on imports of wood

19. A product has a high price elasticity of supply.

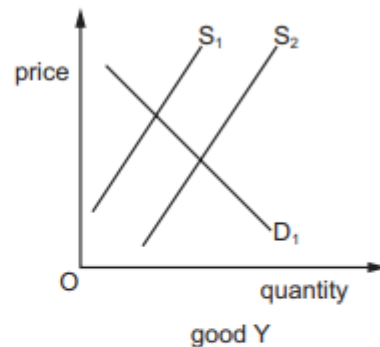
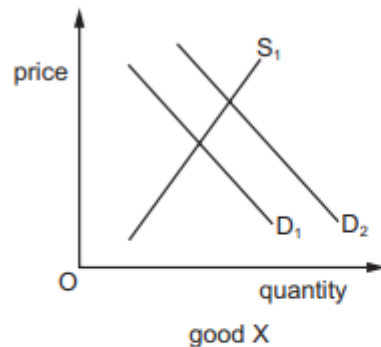
What might explain this?

- A It has a high opportunity cost.
- B It has a non-perishable nature.
- C It is classed as an inferior good.
- D It is classed as a luxury good.

20. What is likely to affect the position of the demand curve for tickets to pop concerts but **not** the position of the supply curve?

- A a change in incomes of potential customers
- B a change in price of tickets to the concerts
- C a change in rents charged by venue owners
- D a change in seating capacity of concert venues

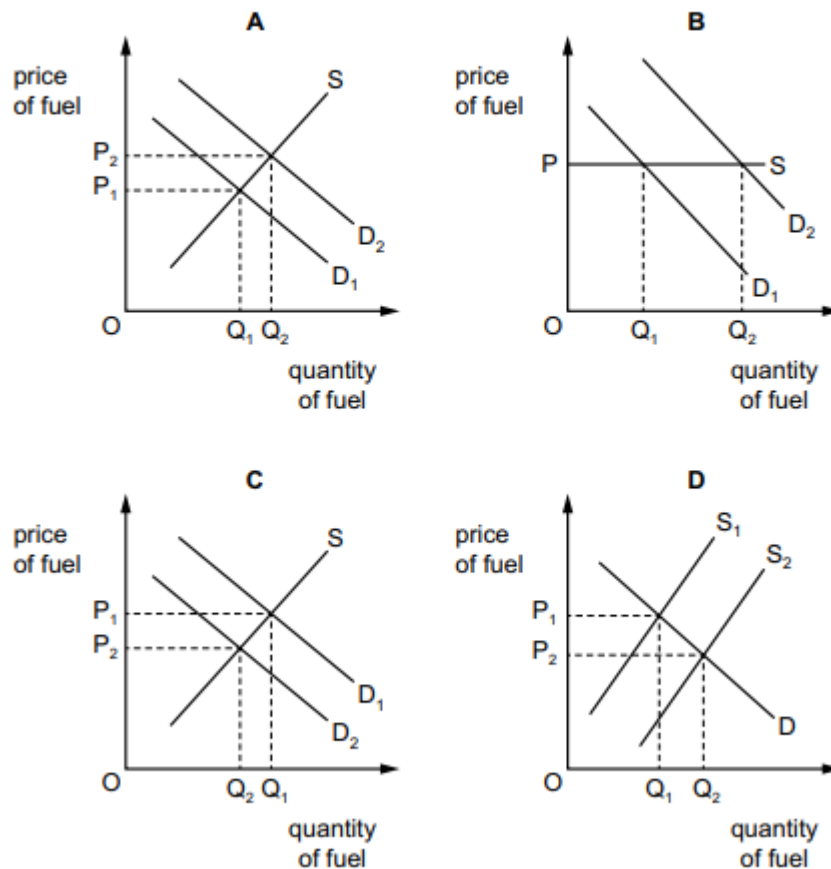
21. The diagrams show the effect of a change in the market for good X on the market for good Y.



What can be concluded about the goods?

- A X and Y are complements.
- B X and Y are in joint supply.
- C X and Y are substitutes.
- D Y is a derived demand from X.

22. What causes an inelastic market supply curve for an agricultural crop such as wheat?
- A a government decision to import crops at times of shortage
 - B an efficient use of fertilisers causing crop production to rise
 - C the storage of excess production for future sale
 - D the very long time required to produce additional output
23. Which statement about price elasticity of supply is correct?
- A It cannot change in the short run.
 - B It increases as the time period lengthens.
 - C It is infinite in the momentary time period.
 - D It is zero in the long run.
24. In 2016 car drivers bought more fuel because the price of oil from which it was made had fallen.
- Which diagram represents this change in the market for fuel?



25. In an attempt to reduce crime, a government introduces a subsidy on the production of burglar alarms, but sales rise very little.

Which possible characteristics of the good would lead to this outcome?

	price elasticity of demand	price elasticity of supply
A	high	high
B	high	low
C	low	high
D	low	low

26. How might a firm raise the value of the price elasticity of supply for its product?

- A** decrease the amount of labour that it employs
- B** employ new technology to increase its productive capacity
- C** increase advertising expenditure to generate more revenue
- D** reduce the level of stocks in order to cut costs

27. Shoe manufacturers often use leather to make their most expensive shoes. Due to a world shortage of leather, the price of leather has risen.

What will be the effect of this price change?

- A** The demand curve for leather shoes will shift to the left.
- B** The demand curve for a substitute for leather shoes will shift to the right.
- C** The supply curve for leather shoes will shift to the right.
- D** The supply curve for a substitute for leather to make expensive shoes will shift to the right.

28. A good has a price elasticity of supply of 2.0. The current quantity supplied is 300 units per week at a market price of \$20 per unit. The firm raises the price to \$25 per unit.

What will the new quantity supplied be per week?

- A** 150 units
- B** 375 units
- C** 400 units
- D** 450 units

29. A farmer divides his land between growing two crops – wheat and oats. To increase production of one he must reduce production of the other.

What would cause the farmer to increase his supply of wheat?

- A A subsidy is given for the production of oats.
- B A unit tax is imposed on the production of wheat.
- C There is an increase in wages for all farm workers.
- D There is improved efficiency in the harvesting of wheat.

30. What is correct about market supply?

- A Market supply can increase only when all individual firms increase their output.
- B Market supply is effective when consumers have sufficient income to buy the good.
- C Market supply is the result of aggregating the supply of all individual firms.
- D Market supply of an inferior good falls as price increases.

Case Study 1

- (a) **Use a demand and supply diagram to show why there was a 'huge' increase in the price of Arabica coffee beans in 2021.**
- Supply and demand diagram with correct axes and showing shift to the left of the supply curve **(1)**.
 - New equilibrium price and quantity **(1)**.

Guidance: No explanation is required.

- (b) **Is the price elasticity of supply of coffee beans likely to be elastic or inelastic in any one year? Justify your answer from the information provided.**

Inelastic **(1)** (accept perfectly inelastic)

Reference to any one of the climatic problems referred to in the text such as drought and frosts in Brazil or storms in Vietnam meaning that supply is volatile/unpredictable. The candidate may refer to the nature of agricultural goods meaning that you cannot simply grow additional output.

The explanation must refer to how it affects the responsiveness of supply **(1)**.

- (c) **Assess the extent to which coffee bean producers will gain from the huge increase in coffee bean prices in 2021.**

Up to 3 marks for explanation/analysis of the gains to coffee bean producers:

- If the demand for coffee is price inelastic in demand, then coffee producers would be expected to gain (in terms of increased revenue).
- For example, increased revenue per lb. of coffee sold; increased funds available to invest and innovate in better methods of storing coffee beans in the event of future storms.
- Producers in Brazil and Vietnam and with existing stocks of coffee that survived the frost and storms would expect to increase their own sales at the expense of those rivals unable to produce a large enough yield of coffee.
- Similarly, producers in countries not affected by climatic problems should benefit from increased sales and revenue.

Up to 3 marks for explanation/analysis of the extent to which coffee bean producers will not gain:

- If the demand for coffee is price elastic in demand, then coffee producers would not be expected to gain (in terms of reduced revenue).
- For example, those producers unable to produce a sufficiently large enough yield may make losses which may ultimately result in business closures.
- Similarly, depending on the strength of the XED, total revenue may fall if the huge price increase in 2021 results in sufficient numbers of consumers switching their preferences to substitute goods e.g. tea.

Up to 3 marks maximum for each perspective, with an overall maximum of 4 marks.

Up to 2 marks for evaluation:

- That clearly considers whether or not coffee bean producers will gain from the huge increase in coffee bean prices in 2021.
- Comes to a reasoned conclusion as to whether advantages outweigh disadvantages or vice versa (reserve 1 mark)

Note: No mark for eval can be awarded if only one perspective considered.

Case Study 2

- (a) **Using Fig. 1.1, calculate the percentage change in the world price of olive oil between May 2022 and January 2023.**

An increase of 43.8% (2); award one mark if no percentage is shown/increase is not stated.

- (b) **With reference to Fig 1.2 and the help of a demand and supply diagram, consider the possible impact of increased olive oil prices in 2023 on consumer surplus.**

Correctly labelled diagram showing initial equilibrium. (1)

Leftward shift in Supply curve establishing new equilibrium at a lower P and increased output. (1)

Explanation of changes with clear reference to diagram provided. Correct shift (1).

The extent of the change in consumer surplus will depend on the PED (1)

Use textbox to show mark distribution. (2,1)

- (c) **Assess the likely impact of increasing stocks of olive oil on the price and quantity of olive oil traded in the future.**

For explanation/ analysis (up to 4 marks)

For an explanation that the sales of stocks will increase supply **(1)** and reduce equilibrium price **(1)** when olive harvests are low **(1)**. This will reduce volatility in the market **(1)** and stabilize incomes for producers **(1)**.

However, difficulties include the necessary the cost of storing the olives, the need to keep a sizeable stock to make a difference to olive prices, the opportunity cost of using/renting land in this way.

Evaluation

That clearly assesses the likely impact on both the price and quantity of olive oil e.g. by considering the extent of any price change being dependent upon that which sufficient stocks can be held – **reserve on 1 mark for a justified conclusion.**

Please use a text box to show the mark split e.g., 3,1

SECTION B B

Question	Answer	Marks
1	With the help of a diagram, explain what could cause an increase in demand for a product <u>and</u> consider whether the impact of an increase in demand on the price of the product will be the same in the short run and the long run. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and understanding (max 3 marks) An understanding of the demand for a product, shown through an accurate diagram, with correctly labelled axes (1), correctly labelled demand and supply curves and a correctly labelled shift of the demand curve to the right (1) and with equilibrium positions in terms of both price and quantity clearly shown or a movement from E1 to E2. (1) AO2 Analysis (max 3 marks) Uses the diagram to explain what could cause a shift of the demand curve to the right for a product and analyses the causes of an increase in demand for a product. AO3 Evaluation (max 2 marks) Offers a valid judgement on whether the impact of an increase in demand on the price of the product will be the same in the short run and long run (1) to reach a conclusion. (1)	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
2	Assess whether the concept of price elasticity of supply or cross elasticity of demand will be the more useful to a business wanting to increase its total sales in a growing economy. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis Analysis of the concept of price elasticity of supply may explain its use in increasing a business' total sales by examining the following. • Usefulness depends on the type of goods and whether decisions need to be made in the short run or long run. • It may help businesses with regards to stock holding and decisions about which products to focus on • It may help businesses make decisions about productive capacity and whether e.g., to invest in new technology BUT • It depends on the accuracy of the data used to produce the value. • This may change over time and the value may have little value in a growing economy. • Some products, such as agricultural/perishable products, will naturally have a more price inelastic supply and knowledge of the value may have little use. Analysis of the concept of cross elasticity of demand may explain its use in increasing a business's total revenue by examining the following. • It will mean the business understands the relationship between its product and others i.e., whether they are complements or substitutes. • It also measures the closeness of the relationship and informs decisions about how quickly and the extent to which it should react for example if a competitor changes its prices. • It should assist a business to develop a pricing strategy that maximises opportunities to generate more sales BUT • The usefulness depends on the accuracy of the data and in a growing economy, it may not be possible to keep the value up to date. It is therefore often dependent on estimates that may be inaccurate. • Businesses may not be able to act on the data by changing prices if their cost structure cannot be easily changed. • Other variables other than cross elasticity of demand may be more important for example non price determinants such as brand loyalty. Better answers will refer directly/implicitly to their usefulness to a business operating in a growing economy. Answers which do not are unlikely to gain more than L3 lower band. One sided response i.e., those that only consider one concept or do not analyse the problems of at least one concept will not gain beyond L2.	12

Question	Answer	Marks
2	AO3 Evaluation • Consideration of the likely effectiveness of both measures in increasing total sales in a growing economy that assesses which is most likely to be effective. If there is no direct/implicit reference to a growing economy, then L1 upper maximum. • Candidates may consider an alternative concept e.g., YED to assess whether sales would actually grow in these circumstances, • A justified conclusion which is most likely to be effective. Accept all valid responses. A one-sided response cannot gain any marks for evaluation.	
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
3	The price elasticity of supply (PES) for a new smartphone is estimated to be 0.8 in the short run and 1.8 in the long run. Explain what these estimates mean for producers and consumers of smartphones <u>and</u> consider why the estimates differ. Follow the point-based marking guidance at the top of this mark scheme and award: - up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis - up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) Meaning of price elasticity of supply (1) an accurate formula (1) an explanation of what is meant by elastic/inelastic based on the responsiveness of changes in supply to changes in price (1). AO2 Analysis (max 3 marks) Analysis of why the values may differ due to e.g., the time period, level of stock, changes in the production process etc. Up to 2 marks for each reason explained. Maximum of 3 marks overall AO3 Evaluation (max 2 marks) An assessment as to why the impact on consumers and producers may differ according to these different estimates e.g., on availability, pricing and investment decisions.	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
4	Assess whether the supply of agricultural products is likely to be more price elastic or less price elastic than the supply of manufactured products. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis A knowledge and understanding and analysis of possible factors affecting the PES of agricultural products and manufactured products, including: • the number of producers • the existence of spare capacity • the ease of storing stocks • the time period • the extent of factor mobility • the length of the production process. Answers should analyse how PES is affected by each of these, with a clear distinction between the two types of products. Level 1 responses will be assertive and lacking in explanations/mainly descriptive and/or mainly lacking in relevance to the question. Level 2 responses may contain some inaccuracies and will be one-sided. Analysis will be explained at least in part and will be largely relevant to the question. Level 3 responses will consider alternative policies/concepts etc. and will be balanced. Explanations of points raised will be offered and will be accurate and relevant to the question. AO3 Evaluation • Evaluation of whether the supply of agricultural products is likely to be more price elastic or less price elastic than the supply of manufactured products. • In conclusion, PES for agricultural products is likely to be less elastic than the PES for manufactured products. Accept all valid responses. A one-sided response cannot gain any marks for evaluation.	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
5	Assess whether an estimate of the price elasticity of demand for a product is likely to be more useful to a firm than an estimate of its price elasticity of supply. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis • The meaning of the terms PED/PES • How PED estimates would be useful to business owners –it would indicate the likely changes to revenue in response to changes in the price of a product. • An explanation of the limitations of PED estimates – does not consider the prices of complements/substitutes; income level of the consumer; trends; ignores the relevance of costs so changes to profits are uncertain. • An explanation of how PES estimates would be useful to business owners - the potential to invest in spare capacity which can be used if demand rises; increase overtime payments to increase production; whether or not to outsource to other firms who can meet supply. • An explanation of the limitations of PES estimates – e.g., practical ability to increase the PES; the associated costs in making supply more elastic/responsive. AO3 Evaluation • Consideration that fully assesses the relative effectiveness of both PED/PES estimates and their implications for business decision-making and arrives at a reasoned conclusion concerning which concept (or otherwise) is most useful. Accept all valid responses.	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
6	Assess the extent to which price elasticity of supply or cross elasticity of demand is most useful to businesses. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis - An assessment of the usefulness of PES e.g., it estimates how quickly a business will take to respond to changes in demand, levels of stock it can hold etc., but this is only an estimate and may change. - An assessment of the usefulness of XED e.g., it tells the business whether their product is a substitute (positive XED) or a complement (negative XED) and how close the relationship is and hence its sensitivity to the actions of other businesses, but the speed of change may mean it is quickly out of date. Level 1 responses will be assertive and lacking in explanations / mainly descriptive and/or or mainly lacking in relevance to the question. Level 2 responses may contain some inaccuracies and will be one sided. Analysis will be explained at least in part and will be largely relevant to the question. Level 3 responses will consider alternative policies / concepts etc. and will be balanced. Explanations of points raised will be offered and will be accurate and relevant to the question. AO3 Evaluation - For a full evaluation of the usefulness of both measurements compared with each other leading to: - A valid conclusion Accept all valid responses.	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

Question	Answer	Marks
7	With the help of a formula, explain the meaning of price elasticity of supply <u>and</u> consider which factors are most important in determining whether the price elasticity of supply for a good is likely to be relatively elastic or relatively inelastic. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and understanding (max 3 marks) What the price elasticity of supply measures (1) and the formula for price elasticity (1) and the distinction between relatively elastic and relatively inelastic. (1) AO2 Analysis (max 3 marks) Up to 2 marks for explaining each factor that makes elasticity of supply elastic/inelastic. (max 3 marks) Guidance: 2 marks for each well-explained factor and 1 mark for each factor with limited explanation. (max 3 marks) AO3 Evaluation (max 2 marks) Offers a valid judgement on the relative importance of the factors that influence the price elasticity of supply of a product (1) to reach a conclusion. (1)	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
8	Assess the likely success of any <u>two</u> government policies to make the price elasticity of supply of all agricultural products more elastic. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 8 marks. AO3 out of 4 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis How two government policies can influence elasticity of supply of agricultural products. These might include: • buffer stocks • provision of stocks/warehousing/refrigeration • improved infrastructure in the agricultural sector • training of agricultural workers • agricultural subsidies. Explanation of the advantages and disadvantages associated with each measure. Explanation of why the measures might be ineffective. AO3 Evaluation • Consideration of the relative effectiveness of the two policies explained. • Consideration of the impact of these policies on different types of agricultural products. • In conclusion, whether the policies will be effective for all agricultural products. Accept all valid responses.	12
	AO1 Knowledge and understanding and AO2 Analysis	8
	AO3 Evaluation	4

SECTION C

1.	A	11.	C	21.	B
2.	B	12.	A	22.	D
3.	C	13.	B	23.	B
4.	C	14.	C	24.	D
5.	C	15.	A	25.	D
6.	C	16.	C	26.	B
7.	D	17.	C	27.	B
8.	A	18.	C	28.	D
9.	A	19.	B	29.	D
10.	B	20.	A	30.	C